

COURSE TITLE : YARN FORMATION III
COURSE CODE : 5062
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Modern Developments in Ring Spinning	13
2	Advanced Yarn Manufacture	13
3	Doubling, Fancy yarn and Reeling	13
4	Calculation	13
TOTAL		52

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Understand various developments in ring frames.
	2	Know the features of compact spinning
2	1	Know the modern techniques of yarn formation by rotor
	2	Understand the features of friction, airjet and siro spinning
3	1	Understand different yarn doubling techniques, ring doubler & TFO
	2	Understand the features of fancy doubling & reeling
4	1	Understand the draft, twist, production and efficiency of Ring frame
	2	Know production of rotor, ring doubler & TFO

SPECIFIC OUTCOME

MODULE I MODERN DEVELOPMENTS IN RING SPINNING

1.1.0 Understand various developments in ring frames

- 1.1.0 Developments in drafting region
- 1.1.1 Explain the features of INA-V drafting system
- 1.1.2 Developments in twisting region
- 1.1.3 Define Traveller number
- 1.1.4 Design features of Energy saving spindle
- 1.1.5 Variable speed spinning
- 1.1.6 List Features of auto doffing mechanism
- 1.1.7 Advantages of long length frames
- 1.1.8 explain the super high drafting system

1.2.0 Know the features of compact spinning

- 1.2.1 State the Importance of compacting element to ring frame
- 1.2.2 Explain the elite compacting element
- 1.2.3 State the function of different parts in elite compact spinning.
- 1.2.4 Explain the Advantages of compact spun yarns

MODULE II ADVANCED YARN MANUFACTURE

2.1.0 Know the modern techniques of yarn formation by rotor

- 2.1.1 List the different advanced yarn manufacturing systems
- 2.1.2 State Principle of Rotor spinning.
- 2.1.3 Explain the passage of material through a rotor spinning machine
- 2.1.4 State the functions and detailed study of various parts
- 2.1.5 Explain the effect of opening roller speed on yarn quality
- 2.1.6 List the types of rotor grooves
- 2.1.7 Describe effect of rotor speed on yarn quality
- 2.1.8 State the Relation between rotor speed, rotor diameter and yarn draw-off tension
- 2.1.9 List the advantages and limitations of open end spinning over ring spinning

2.2.0 Understand the features of friction, airjet and siro spinning

- 2.2.1 List the Principle of friction spinning
- 2.2.2 Explain the material passage and yarn formation in DREF – I friction spinner
- 2.2.3 Describe the material passage and yarn formation in DREF –II friction spinner
- 2.2.4 List the advantages of DREF - II over DREF - I
- 2.2.5 State the Principle of air jet spinning
- 2.2.6 Describe the yarn formation in Murata air jet spinner
- 2.2.7 Explain the working of Siro spinning
- 2.2.8 Differentiate between Siro spinning and Ring spinning
- 2.2.9 list the advantages of Siro spinning

MODULE III DOUBLING, FANCY YARN AND REELING

3.1.0 Understand different yarn doubling techniques, ring doubler & TFO

- 3.1.1 State the objectives of doubling
- 3.1.2 List the types of doubled yarn
- 3.1.3 State the Twist and twist direction effects
- 3.1.4 Explain the dry and wet doubling
- 3.1.5 Compare dry and wet doubling
- 3.1.6 Explain various Styles of doubling
- 3.1.7 List the types of doubling machine
- 3.1.8 Explain the working of Ring doubler
- 3.1.9 Describe the working of Two for one twister-TFO
- 3.1.10 Explain the working of Three for one twister
- 3.1.11 Describe the Manufacture of Sewing threads and Tyre cord

3.2.0 Understand the features of fancy doubling & reeling

- 3.2.1 List the types of fancy yarn
- 3.2.2 Explain the working of a fancy doublers
- 3.2.3 State the objectives of yarn conditioning
- 3.2.4 State the objectives of reeling
- 3.2.5 List the Types of reeling
- 3.2.6 Explain the working of plain reel and cross reel
- 3.2.7 State the Objectives of bundling and baling

MODULE IV CALCULATIONS

4.1.0 Understand the draft, twist, production and efficiency of Ring frame.

- 4.1.1 Draw the gearing diagram of Ring frame
- 4.1.2 Determine Ring frame Drafts
- 4.1.3 Determine the draft and draft constant
- 4.1.4 Estimate the twist and twist constant of Ring frame
- 4.1.5 Find the Production and efficiency of the ring frame
- 4.1.6 State the Definition of angle of yarn pull
- 4.1.7 Estimate the Angle of yarn pull
- 4.1.8 Find the Traveller speed

4.2.0 Know production of rotor, ring doubler & TFO

- 4.2.1 Determine draft and twist in rotor spinning
- 4.2.2 Find the Production and efficiency of the rotor spinning
- 4.2.3 Estimate production and efficiency of Ring doublers
- 4.2.4 Determine the production and efficiency of Two for One twister

CONTENT DETAILS

MODULE I MODERN DEVELOPMENTS IN RING SPINNING

Developments in drafting region - Stepped nose bar- opposite helically fluted bottom drafting rollers- INA-V drafting system-Developments in twisting region-traversing thread guide – anti-wedge ring and elliptical traveller- ABC rings – antifriction plug type spindles – energy saving spindles -Define Traveller number-Variable speed spinning – Dual motor drive – variable speed motor drive – VPS drive – Micro processor controlled Various Speed drive - Features of auto doffing mechanism - long length frames - super high drafting system – feed requirements and the drafting arrangement of super high drafting system .

Compact spinning

Objectives of compacting element – spinning triangle - Elite compacting element – elite top roller – lattice apron – profile tube – function of vacuum - Advantages of compact spun yarns.

MODULE II ADVANCED YARN MANUFACTURE

Different advanced yarn manufacturing systems –Rotor – Friction – Air Jet & Siro Spinning.

Rotor Spinning

Principle of Rotor spinning - yarn formation in rotor spinning machine – feed part – opening roller – transport channel - rotor – navel – take up mechanism - opening roller speed and yarn quality - types of rotor grooves –S groove , V groove, U groove , T groove, G groove - yarn characteristics of different types of rotors - rotor speed and yarn quality- rotor diameter and rotor speed – fibre length and rotor diameter- Relation between rotor speed, rotor diameter and yarn draw-off tension - advantages and limitations of open end spinning over ring spinning.

Friction, airjet and siro spinning

Principle of friction spinning - DREF – I friction spinner - DREF –II friction spinner- Principle of air jet spinning - Murata air jet spinner – Fasciated yarn - Siro spinning - compare Siro spinning and Ring spinning - compare Siro spinning and ring doubler .

MODULE III DOUBLING, FANCY YARN AND REELING

Objectives of doubling – Types of doubling machine – Ring doubler – Two- for- one twister – Three – for-one twister - Objectives of doubling - types of doubled yarn – Ply , cord and cable yarn - Twist and twist direction effects – balanced and unbalanced yarn - dry and wet doubling - Styles of doubling – English style, Stockport style, modified Stockport style, American style, Scotch style and Nottingham style - Ring doubler, stop motion - Two-for-one twister- Three-for-one twister - Sewing threads and Tyre cord.

Fancy doubling & reeling

Types Fancy yarn – spot yarn, loop yarn, gimp yarn - fancy doubler- yarn conditioning – tank system , vapour system , trough and roller system, spraying system - objectives of reeling - Types of reeling- plain reel and cross reel - Objectives of bundling and baling.

MODULE IV CALCULATION

Ring frame - draft, twist, production and efficiency calculation.

Gearing diagram of Ring frame- draft- break draft constant and main draft constant – twist multiplier (TM) - twist and twist constant – TM for hosiery , medium and high twist yarns - Production and efficiency of the ring frame , spindle point production – Yarn realization - angle of yarn pull – relation between angle of yarn pull ,bobbin diameter and ring diameter -traveller speed – relation between ring diameter , spindle speed and traveller speed – maximum practical traveller speed.

Production of rotor, ring doubler& TFO

Draft and twist in rotor – production –efficiency - production of Ring doubler, TPI, resultant count , average count - production and efficiency of Two for One twister

REFERENCE BOOKS

- | | |
|---|--------------------------------|
| 1. Taggart - Cotton spinning | – Mahajan Publishers, Amhdabad |
| 2. Pattabiraman - Essential elements of Cotton Spinning - | – Mahajan Publishers, Amhdabad |
| 3. G Merril - Cotton Ring spinning | – Mahajan Publishers, Amhdabad |
| 4. RV.Mahendra Gowda - New Spinning systems | - NCUTE |
| 5 Nield - Open end spinning | – Mahajan Publishers, Amhdabad |
| 6 Taggart - Cotton Spinning Calculation | – Mahajan Publishers, Amhdabad |